Hamza Alshamy

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Education

New York University

M.S. in Data Science (GPA: 3.9/4) — May 2026
- Research Interests: Human-AI Interaction; Sociotechnical AI; Computational Social Science

B.A. in Economics — May 2024
- Minors: Data Science and Mathematics

Research Experience

Center for Conflict and Cooperation — Aug 2025 – *Present*
Research Assistant – Human-AI Interaction — New York, NY
PI: Dr. Jay Van Bavel | Post-doc: Dr. Laura Globig

- **Social media incentives & harmful behavior:** Developed an NLP pipeline (Perspective API, PCA) to measure online content harmfulness across 10K+ posts; validated against a published benchmark ($r = 0.992$) and packaged parameters for large-scale inference.
- Designed simulated social feedback grounded in observed human behavior, using survey data to parameterize a large-scale online experiment.
- **AI vs. human information seeking:** Studied how people choose between AI and human information sources across three experiments ($n \approx 1{,}600$); implemented hierarchical drift-diffusion models to characterize how source identity shapes evidence accumulation during deliberation.

Kensho Technologies (S&P Global) — Sep 2025 – Jun 2026
Student Researcher – Natural Language Processing — Cambridge, MA (Remote)
Supervisors: Dr. Chris Tanner & Dr. Craig Schmidt

- Built a tokenizer that learns its vocabulary by directly optimizing compression, formulating the problem as an integer program solved at corpus scale with Gurobi.
- Benchmarked learned vocabularies against BPE and a certified optimality bound, closing most of the remaining compression gap and quantifying how near-optimal standard tokenizers are.

Centre for Human-Inspired AI – University of Cambridge — Jul 2025 – May 2026
Visiting Student Researcher – Human-AI Interaction — Cambridge, UK (Remote)
PI: Dr. Umang Bhatt | Lab: Trustworthy AI Lab

- Co-authored STaCK, a framework for mapping the developmental risks of AI deployment across career stages and identifying tasks where automation may displace opportunities for skill development.
- Built a co-occurrence network of 64 software engineering sub-tasks from 123K+ job postings; designed and applied Leiden community detection and centrality analysis to identify task communities and developmental hubs.

King Faisal Center for Research and Islamic Studies — Sep 2024 – May 2026
Contracted Researcher – Computational Social Science — Riyadh, SA (Hybrid)

- Across projects, used computational methods to study Saudi Arabia's economy and society, from international media portrayals to demographic change.

New York University — Dec 2023 – May 2024
Undergraduate Quantitative Researcher — New York, NY
Faculty Advisor: Dr. Gerald McIntyre

- Built and cleaned a 100-country World Bank panel dataset, and applied fixed and random effects models to examine demographic predictors of GDP growth and unemployment.

Employment

Subconscious AI — Oct 2026 – *Present*
Founding Researcher, Simulations & Evaluations — New York, NY

• Leading the evaluations function for LLM-based simulations of human behavior. Measuring where simulations are reliable, where they fall short, and how to improve them.
• Designing benchmarks that compare simulated and human responses across domains, drawing on conjoint studies, the General Social Survey, and surveys fielded after model training cutoffs.
• Drawing on Computer Science, Economics, and Psychology to shape research methodology, including structured experiment design and baselines set before model changes.

Working Papers

[1] **Artificial Intelligence Circumvents Identity-Driven Biases in Source Selection**
Under Review at Computers in Human Behavior
Laura K. Globig, Rachel Xu, **Hamza Alshamy**, and Jay Van Bavel

[2] **STaCK: Mapping and Scoring the Developmental Risk of AI Deployment Across a Career Ladder**
Under Review
Noah Broestl, **Hamza Alshamy**, Umang Bhatt, Adrian Weller

[3] **Joint Optimization for Greedy Longest-match Tokenization**
Tokenization Workshop (TokShop) at Conference on Language Modeling (COLM) 2026
Adhiraj Singh, Deepanshu Mody, Ghina AlShdaifat, **Hamza Alshamy**, et al.

[4] **Using Reward-Learning to Reduce the Spread and Impact of Harmful Online Content**
In Progress
Laura K. Globig, **Hamza Alshamy**, Jay Van Bavel

[5] **Animated Territorial Data Extractor (ATDE): A Computer-Vision Method for Extracting Territorial Data from Animated Historical Maps**
arXiv Preprint (2025)
Hamza Alshamy, Isaiah Woram, Advay Mishra, Zihan Xia, Pascal Wallisch

Fellowships, Funding, & Awards

King Abdullah Full Merit Scholarship – Graduate Studies — 2024 – 2026

Frédéric Bastiat Fellow – Economics — 2024 – 2025
The Mercatus Center at George Mason University

Best Social Science Research Award — 2024
50th Annual Undergraduate Research Conference, NYU

NYU Dean's Undergraduate Research Fund — 2024

Joseph Schumpeter Fellow – Economics — 2021 – 2022
The Mercatus Center at George Mason University

King Abdullah Full Merit Scholarship – Undergraduate Studies — 2020 – 2024

Teaching Experience

New York University, Graduate Teaching Assistant

DS-GA 1001: Introduction to Data Science (*Graduate Course*) — *Fall 2025*
Instructor: Dr. Pascal Wallisch

DS-UA 201: Causal Inference — *Fall 2025*
Instructor: Dr. Sidharth Sah

CSCI-UA 473: Fundamentals of Machine Learning — *Spring 2025*
Instructor: Dr. Pascal Wallisch

DS-UA 112: Principles of Data Science II — *Spring 2025, Spring 2026*
Instructor: Dr. Pascal Wallisch

Select Projects

A more exhaustive list of my projects can be found on GitHub.

Trees: Interactive Visualizer for Tree-Based Machine Learning Algorithms

• Built an interactive web application to visualize and step through training and inference of decision trees, bagging, and boosting models in real time.

• Implemented CART from scratch (classification and regression) using Gini impurity and MSE, with bootstrap aggregation and out-of-bag evaluation.

Large-Scale Movie Recommender System

• Built a recommender on the full MovieLens dataset (33M ratings) with Apache Spark on NYU's HPC cluster. Combined LSH-based user similarity with a tuned ALS model, evaluated on MAP@k and NDCG@k against a popularity baseline using a per-user chronological split to prevent leakage.

Research Reports & Policy Publications

[1] **The Perception of Saudi Investment Landscape in Western Eyes: An NLP-Based Sentiment Analysis**
Yahya Alshamy* and **Hamza Alshamy***
Internal Government Research Report, 2026

[2] **Saudi Demographic Economics: Population Aging and Drivers of Declining Fertility Rates**
Hamza Alshamy
King Faisal Center for Research and Islamic Studies, Masarat, December 2025

**Equal contribution*

Technical Skills

• **Programming:** Python, R, SQL
• **ML/DL Libraries:** PyTorch, Hugging Face, Transformers, scikit-learn
• **Tools & Frameworks:** Git, HPC, Qualtrics, Overleaf
• **Applied Skills:** Machine Learning, NLP, Experimental Design, A/B Testing, Causal Inference, Probabilistic Modeling, Data Visualization

Select Coursework

Machine Learning (Kyunghyun Cho); Natural Language Understanding (Tal Linzen); Probability & Statistics for Data Science (Carlos Fernández-Granda); Introduction to Psychology; Introduction to Neuroscience